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Murtaza Pakawala

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Website

Education

Sydney, Australia	UNSW	Feb 2022 - Dec 2025
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- Bachelor of Advanced Computer Science (Honours)
- Engineering WAM: 81
- Coursework: Machine Learning and Data Mining, Computer Vision, DeepLearning, Advance C++, Programming Challenges, Extended Operating Systems

Ann Arbor, USA	University Of Michigan	Jan 2025 - May 2025
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- Exchange Student GPA: 3.7
- Coursework: Linear Algebra, Probability, Ethics in Technology

Research Thesis

	UNSW Sydney	May 2025 – Dec 2025
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- **Thesis:** Inferring Evolutionary Parameters from Phylogenies.
- Developing a simulation-based inference pipeline using **msprime** and **PyTorch** to estimate evolutionary parameters from phylogenetic data.

Personal Projects

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- **C++ Neural Network from Scratch** *GitHub Repository* Designed and implemented a neural network from scratch in C++, leveraging object-oriented programming (OOP) principles to modularize different components. Implemented gradient computation and backpropagation using the chain rule, providing a hands-on understanding of the inner workings of neural networks.
 - **Course Tracker** *GitHub Repository*
Designed and developed a **React Native** application to help university students track course progress and gain insights into their overall and individual marks. Built with **Express.js**, **TypeScript**, and **SQL**, the app won 2nd place overall in the Personal Project Competition and was voted the best app by UNSW students.

Extracurriculars

BLOGS

Blog / Website

- Write short blog posts on topics in mathematics and machine learning to explain concepts clearly and build intuition.

CSESoc Projects

GitHub Repository

- Contributed to the Discord Bot project used by thousands of students in the CSESoc Discord server.
- Successfully migrated the codebase from JavaScript (JS) to TypeScript (TS), improving maintainability and enforcing strong typing.

Skills

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- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
 - **Programming Languages:** C++, C, Java, Python, JavaScript, TypeScript, SQL
 - **Frameworks & Technologies:** React, React Native, Next.js, Express.js, NestJS, PostgreSQL

Awards

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- Second Prize in a personnel project competition organized by CSESOC
 - Highest Voted App prize organized by CSESOC